THE PETROLEUM GEOLOGY OF CAMBODIA

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The petroleum geology of Cambodia has been studied less than has that of the neighbouring countries in Indochina. Nevertheless, the results of recent geological, remote- sensing and limited seismic investigations (the latter on the Cambodian Shelf area of the Gulf of Thailand) have allowed identification of sedimentary basins favourable for hydrocarbons exploration (both on- and offshore), and, consequently, the delineation of prospective areas. Six possibly oil- and gas-bearing basins have been identified, namely: the Siam (northern portion: in the Gulf of Thailand), Tonle Sap, Khorat (southern portion), Preah, Chung and Svairieng Basins.

INTRODUCTION

The petroleum potential of Cambodia is here evaluated on the basis of the results of numerous geological reconnaissance surveys, including:

(i) 1:50,000 geological maps compiled by French geologists of the *Service Géologique de l'Indochine* (Saurin, 1937, 1939, 1944; Gubler, 1939), with explanatory notes to individual sheets; these maps were re-issued with additions to the explanatory sheets by the *Direction Géographique Nationale* in Dalat (southern Vietnam) in 1962.

(ii) a 1:200,000 geological map compiled by Fromaget in 1937. A second edition was produced in 1952 with the collaboration of E. Saurin, and a third edition was edited by Fontaine in 1971.

(iii) geological maps of Cambodia on the scale of 1:2,000,000 consisting of 14 sheets, with explanatory notes to individual sheets, compiled by French geologists of the Bureau de Récherches Géologiques et Minières on behalf of the Khmer Republic, based on the results of field surveys in 1967-1973.

(iv) an unpublished 1:500,000 map of Cambodia prepared by VNIIzarubezhgeologia (VZG) based on satellite remote sensing and aerial photography;

and the results of recent field work by the Authors in the Gulf of Kampong Saom and nearby islands in the Gulf of Thailand, and onshore Tonle Sap and Svairieng Basins.

The geological map of Cambodia (Fig. 1) indicates that rocks of Precambrian to Cenozoic ages are present there (Sauv Chin Hun and Thoch Soval Say, 1986). Hydrocarbons exploration is targeted at Cenozoic prospects on the offshore shelf, and at Upper

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Carboniferous — Cretaceous formations onshore. Prospective areas in Cambodia cover about 150,000 sq. km, and major exploration targets are present in the Cenozoic Rong Depression and the Mesozoic Kampong Saom Trough (both parts of the offshore Siam Basin), and the Tonle Sap Basin.

THE STRATIGRAPHY OF CAMBODIA (Fig. 2)

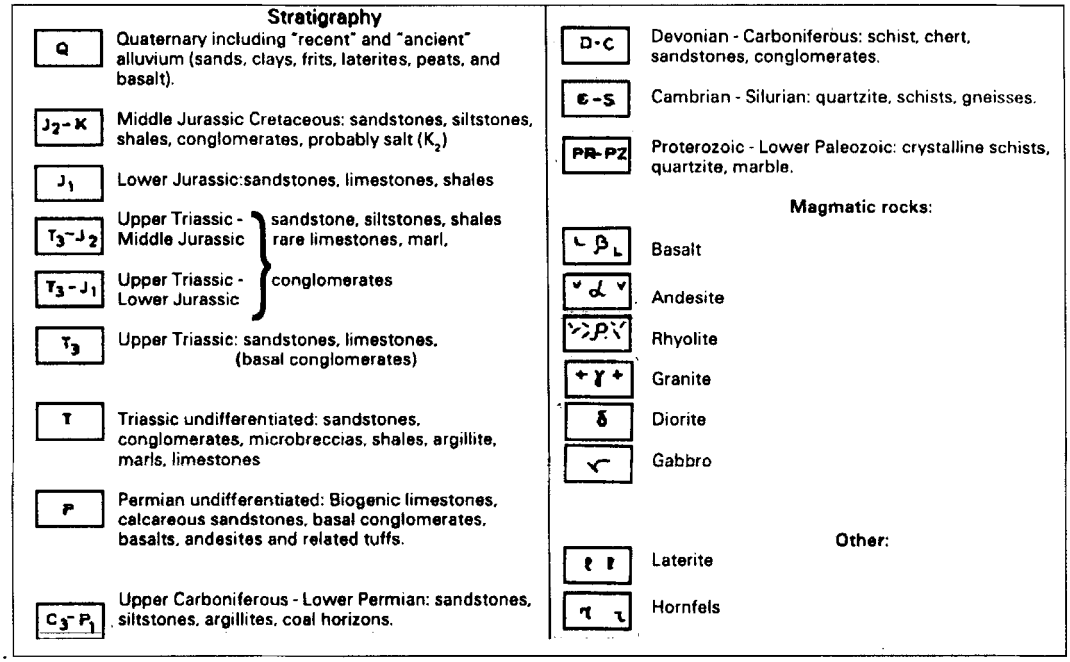
Precambrian — Middle Palaeozoic complexes in Cambodia have been subjected to regional and contact metamorphism, and are cut by numerous intrusions. These poorly-studied complexes form the crystalline or folded basement for large structural depressions, where great thicknesses of low-grade metasediments provide targets for hydrocarbons exploration. In central Vietnam and Laos, island-arc sequences of this age were intensively deformed during the Caledonian Orogeny.

Upper Palaeozoic rocks crop-out mainly in the east of the country. These rocks were studied in detail by Saurin (1937) to the NE of the town of Kracheh. The lower part of the section here is represented by green or greenish-grey shales, interbedded with sandstones; the middle part is composed of black, compact, weakly-bituminous limestones, rich in brachiopods, crinoids and foraminifera; while the upper part consists of oolitic limestones containing *Fusulina alpina* Schellwein and *Fusulina cf. douvillei* Colani dated as Late Carboniferous - Early Permian. The thickness of the succession at outcrop has been estimated at 300 m. Gentle folds have been developed over fault-bounded blocks in the basement, and reactivations of the bounding faults have produced local steep dips.

Permian

Permian rocks outcrop over wide areas in Cambodia. Carbonates are present in the west of the country (in the Pailin and Rumdek Massifs), while carbonate-andesitic successions occur to the east. A carbonate type-section is composed of basal conglomerates, grading upwards into coarse-grained, calcareous sandstones with rare pebbles (derived

Key to Fig. 1 (page 197)



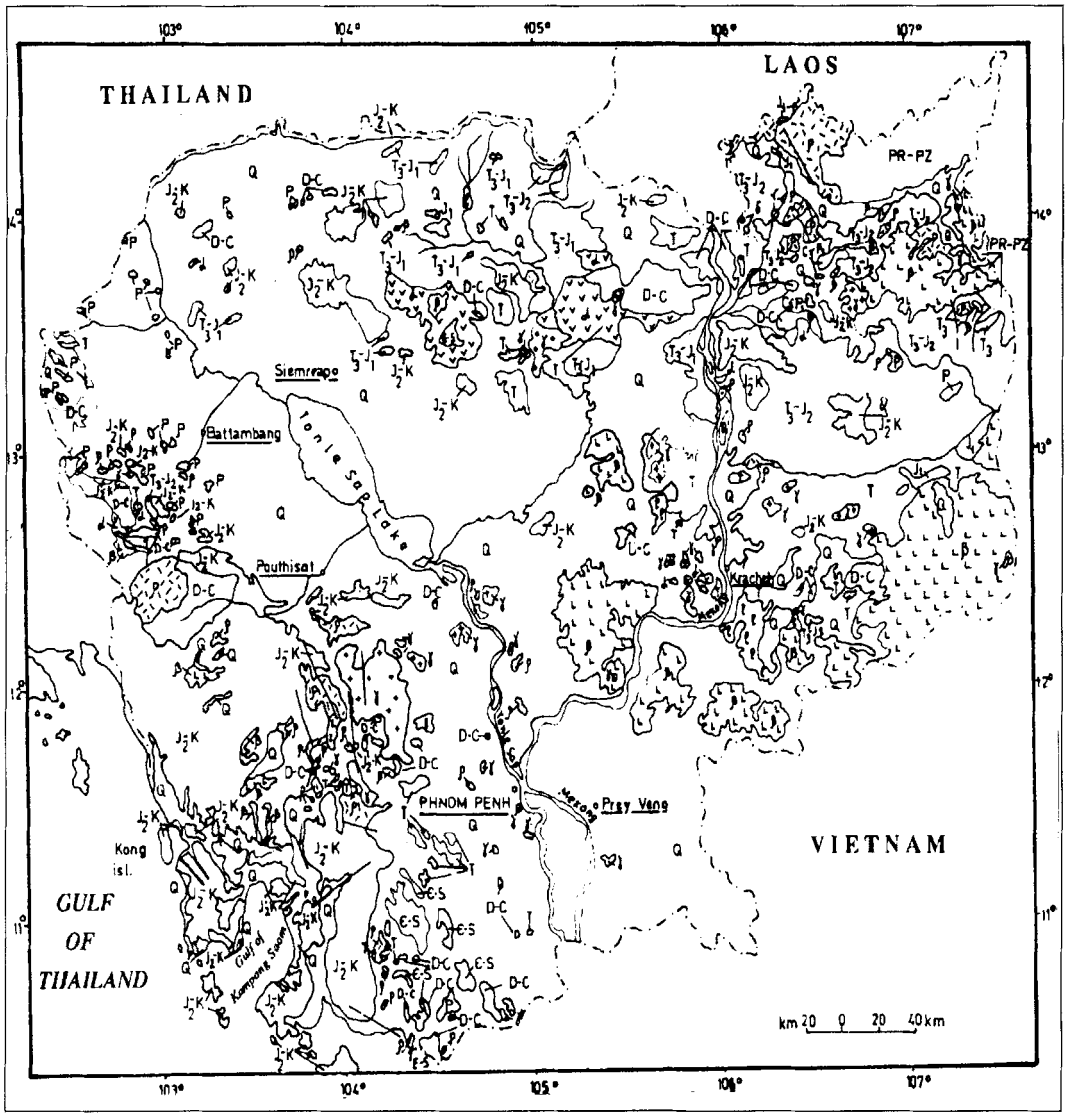


Fig. 1. Geological map of Cambodia.

from the underlying rocks), above which are calcareous sandstones and limestones with abundant fossils. The middle part of the succession is typically composed of organic limestones with abundant and diverse fossils. Clastic material reappears above, where the limestones may frequently contain interbeds of marls, calcareous shales and sandstones.

Permian rocks in the west of Cambodia are estimated to be 400- to 800- m thick. Carbonates are strongly folded, and may be contact-metamorphosed by igneous intrusions into white, coarse-grained, saccharoidal marbles.

Triassic — mid-Jurassic

Permian limestones are overlain by terrigenous rocks of Early – Middle Triassic age. This succession is composed of basal conglomerates, breccias and sandstones, overlain by a marl-shale unit; the upper part is made up of sandstones, shales and conglomerates. The total thickness is estimated to be 1,000-2,000m.

The marl-shale unit is variable, and the shales present in the west of the country are replaced towards the centre by marls and marly limestones interbedded with sandstones, and to the east by shales intercalated with coaly layers. An Early Triassic age has been indicated by the ammonite *Kenodiscus* sp. found in the Rovieng area.

Lower and Middle Triassic rocks are strongly folded and are characterized by medium-grade metamorphism, which corresponds to a vitrinite reflectivity of 1.61-2.55, as determined in coals from the Bantolat area. These rocks are cut by numerous, mainly granitic intrusions, with Indosinian granitoids occupying larger areas than Caledonian and Hercynian intrusives.

Red-coloured terrigenous rocks ("*Terrain rouge*") outcrop as isolated exposures on the flanks of the sedimentary depressions. Their dip angles vary from 7° to 15°. The "*Terrain rouge*" rest unconformably on the basement rocks, and have been dated as Late Triassic to Early — Middle Jurassic; they are deeply buried in the centres of the basins. Fontaine and Workman (1978) considered them to be of lacustrine or lagoonal origin. Fleuriot de Langle (1973) regarded the beds of greyish-blue or greyish-green sandstones as marine formations, alternating with continental sediments in some areas. Marine rocks cropping out along the Srae Pok River are composed of black, fine-grained crystalline limestones that contain *Gryphea arcuata, Oxytoma inequivalva, Pecten hehlii, Spiriferina walcotti* and *Chlamys textorius*.

Late Jurassic — Cretaceous

The "*Terrain rouge*" is overlain by the "*Grès supérieurs*" (i.e. Upper Sandstones, or Upper Indosinian strata) of Late Jurassic — Cretaceous age. These rocks are widespread and fill structural depressions, while outliers form isolated *pnoms*.

The most complete section is exposed to the south of the Elephant Mountains in the Bokor Massif (along the Kdat Creek). Here, the base of the section is composed of siltstones and green clays with gypsum interlayers, which correspond to the "*Terrain rouge*". These are overlain by "*Grès supérieurs*", composed of the following units:

a variegated quartzite-like sandstone with conglomeratic layers (250-300 m thick) at the base;

coarse-grained, clayey sandstones and, more rarely, fine-grained sandstones, together with green clays and siltstones (100-200 m thick);

clayey sandstones and green clays, with rare conglomeratic horizons (150-m thick);

fine-grained, clayey sandstones, with green and wine-red lenses of claystones, interbedded with breccias (100-m thick);

clayey sandstones alternating with massive sandstones, with layers of green clays (200-m thick);

white, quartz arenites with rare interlayers of clay and clayey sandstones (150-m thick).

The total thickness of the sequence is around 1,100 m.

In the middle and upper parts of the Bokor succession and in a clay quarry near the town of Kampong Saom, Beins found *Classopollis* sp., *Diadites spinosa* and *Cicatrico sosporites* (Fontaine and Workman, 1978). These palynological associations are typical of the Lower Cretaceous.

Sandstones on the NW bank of Tonle Sap Lake contain numerous volcanogenic fragments mixed with terrigenous and pyroclastic materials. Volcaniclastic sandstones possess good reservoir properties — for example, effective porosities vary from 18 to 24%. In the Siemreap area, tuffs and tuffaceous breccias are also present.

Cenozoic and Quaternary

Thick sedimentary rocks of Cenozoic age occur on the Cambodian Shelf of the Gulf of Thailand. Onshore, they are represented by thin (mostly Quaternary) formations, and

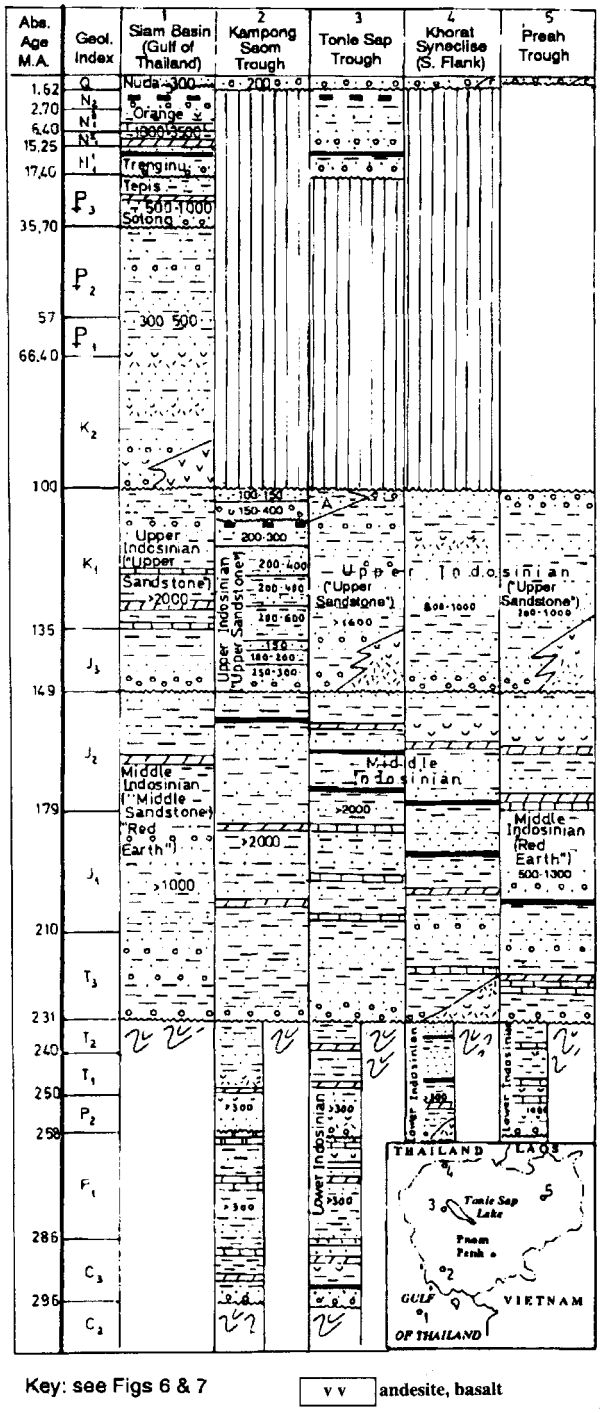


Fig. 2. General stratigraphic column for Cambodia.

only reach significant thicknesses in the Tonle Sap Depression. On the Shelf, the Cenozoic section is composed of some 6,000 m of terrigenous rocks (Fig. 3). Palaeogene strata occur in local graben-like depressions, and are represented by intercalations of marine and continental sandstones, siltstones, shales and coals, with a thickness of up to 2,000-3,000 m.

The most widespread Miocene unit is composed of intercalated sandstones, siltstones and shales, often of deltaic origin. The thickness of this unit ranges from 1,000 m (on uplifts) to 3,000 m (in grabens), and carbonates may be present locally.

Pliocene strata include alternating sandstones, siltstones and clays of marine and continental origin, interbedded with coal and carbonate horizons.

Much of Cambodia is covered by loose sands and clays of Quaternary age, which reach around 250 m in thickness (Carbonnel, 1972). A complete section is present in the Tonle Sap Lake Basin, where "Recent" and "Ancient" alluvia have been identified. Pliocene-Pleistocene flood basalts are also widespread.

LITHOSTRATIGRAPHY AND SEDIMENTATION

Stratigraphic analyses on- and offshore Cambodia have resulted in the identification of the following "structural-formational complexes," which characterise the results of sedimentation in different geodynamic environments:

(i) a C_3-P_1 (Upper Carboniferous — Upper Permian) unit, which is associated with the development of terrigenous-carbonate rocks on the Hercynian shelf;

(ii) a P_2-T_2 (Upper Permian — Middle Triassic) unit, associated with the development of terrigenous-carbonate and volcaniclastic formations in Indosinian rifts and belts of volcanic and plutonic activity;

(iii) a T_3-J_2 (Upper Triassic — Middle Jurassic) unit, associated with the development of transgressive marine and continental rocks in intermontane depressions;

(iv) a J_3-K_2 (Upper Jurassic — Upper Cretaceous) unit, associated with the development of mostly continental terrigenous rocks during the final stages of sedimentation in marginal orogenic depressions;

(v) a KZ (Cenozoic) unit, associated with the development of marine, deltaic, paralic and continental conditions in the intermontane Siam trough, as well as onshore Cambodia.

The palaeogeographical extent of these litho-stratigraphic units is illustrated in Fig. 4. Their structure has influenced processes of hydrocarbons generation, accumulation and preservation.

STRUCTURAL FEATURES

Cambodia is located on the margins of the Indosinian Massif, at the junction of the continental Eurasian, Pacific and Indian blocks (Daly *et al.*, 1991). Continental crust with a thick "granite-gneiss" layer is present over most of the country. This crust is more than 40-km thick in Central Cambodia, decreasing to 25 km on the offshore shelf.

Palaeozoic-Mesozoic fold-belts form three areas of regional uplift (Fig. 5):

(i) The Northern Uplift includes the east-west trending Caledonian Rumbek and Along Chay Massifs, and a portion of the Proterozoic — Early Paleozoic Kontum Massif, located in the extreme NE;

(ii) The Central Uplift includes the Palaeozoic Sam Roung and Plok Massifs;

(iii) The Southern Uplift includes outcropping portions of the Pailin, Preyah and Tasen Massifs. The NE portion of the Tasen Massif adjoins the Hercynian Takongkrao Monocline, which is composed mainly of Upper Carboniferous — Permian limestones.

These Uplifts form the boundaries of a number of Mesozoic sedimentary basins,

(text continued on p. 204)

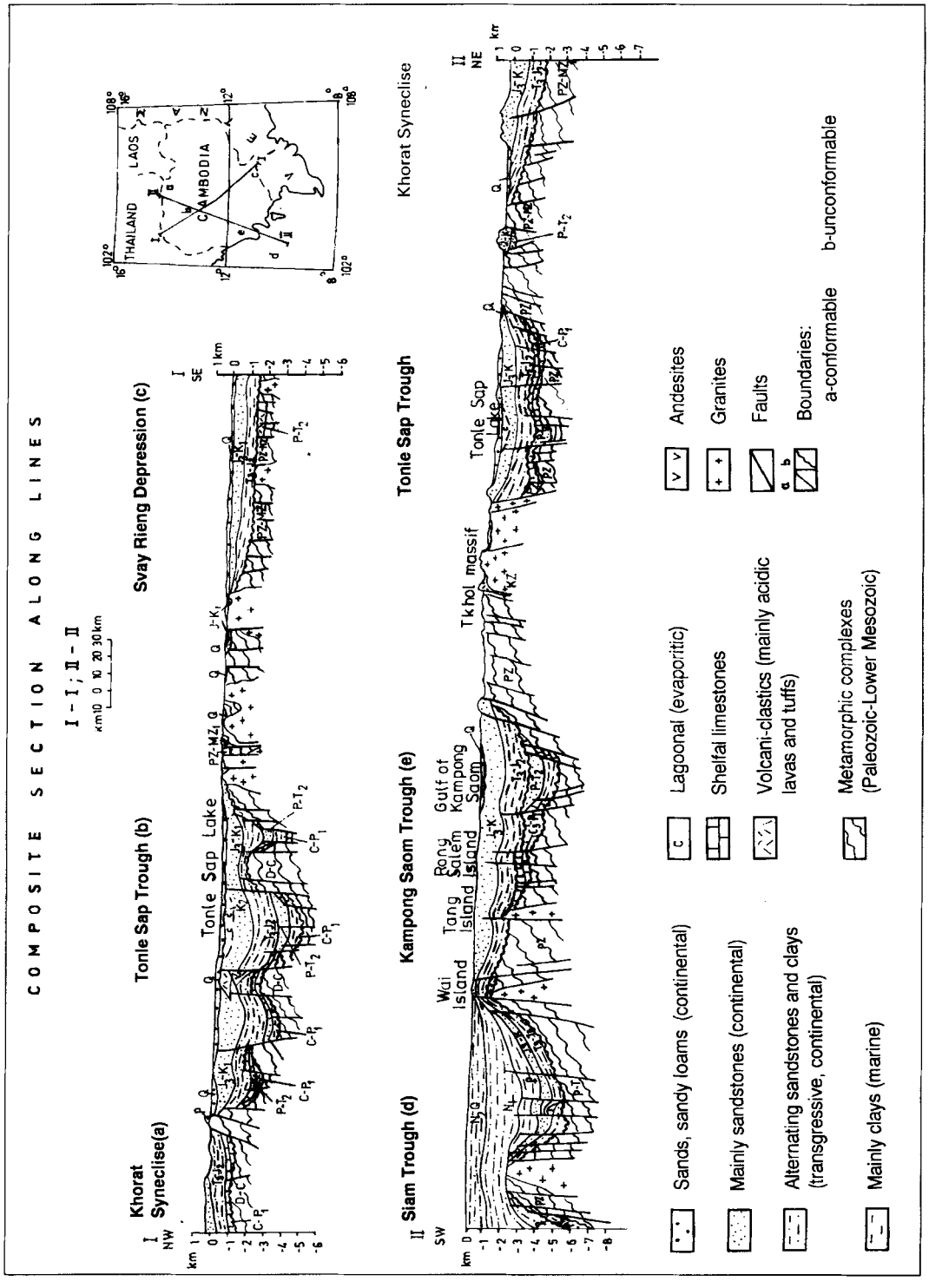


Fig. 3. Composite structural sections: I-I (NW to SE); and II-II (SW to NE).

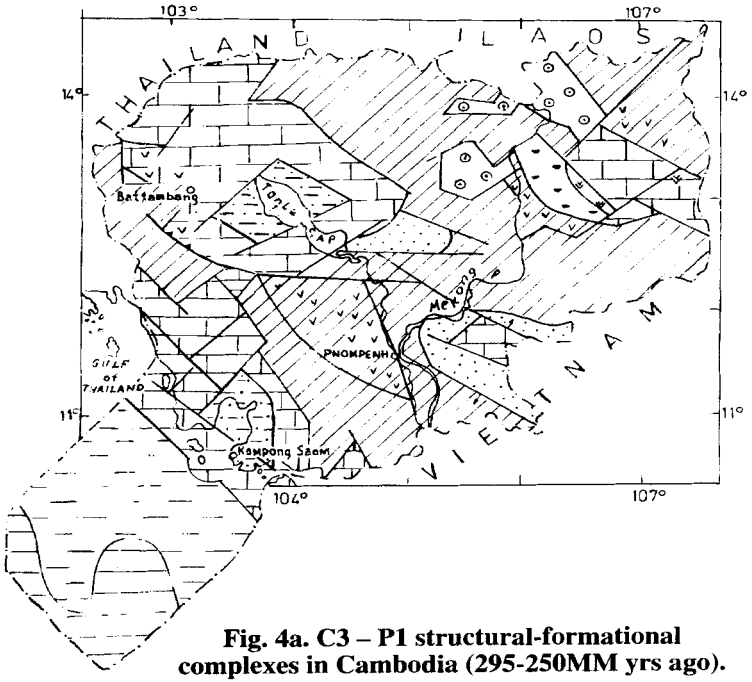


Fig. 4a. C3 – P1 structural-formational complexes in Cambodia (295-250MM yrs ago).

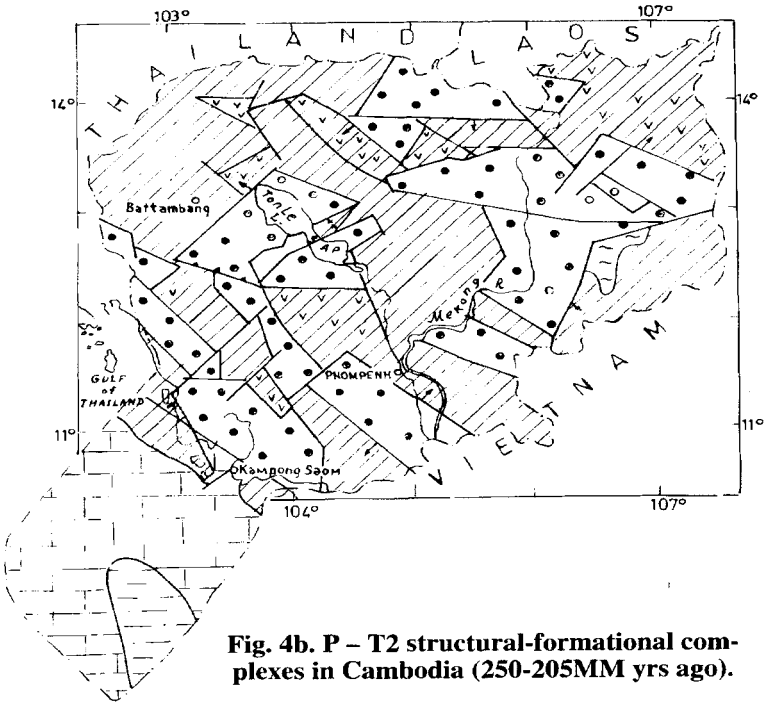


Fig. 4b. P – T2 structural-formational complexes in Cambodia (250-205MM yrs ago).

Structural - formational complexes

- Intracratonal basins.
- Intramontane depressions.
- Marginal portions of intramontane depressions with marine influence.
- Regional elevations in areas of sediment accumulation with interruptions in sedimentation.
- a b — Shelfs (a), including deep shelfs (b).
- Continental slopes and piedmonts.

- Deep-sea troughs.
- "Back-arc troughs" above transitional crust.
- a b c — Continental rifts: (a) which were subsequently inverted; (b) whose development continued subsequently; (c)-Quaternary basalts accompanying the onset of rift formation.
- a b — Volcanic-plutonic "archs" in continental crust, including (a) tectonic-magmatic belts marginal to continental areas; (b) island arcs above transitional crust.
- Elevations within intracratonal and orogenic areas.

Other symbols:

- Boundaries of structural-formational complexes.
- a b — Faults (a), and thrusts (b).
- Inferred directions displacement of lithospheric plates.
- c c — Areas of inferred salt development.
- Areas of maximum subsidence within sedimentary basins.

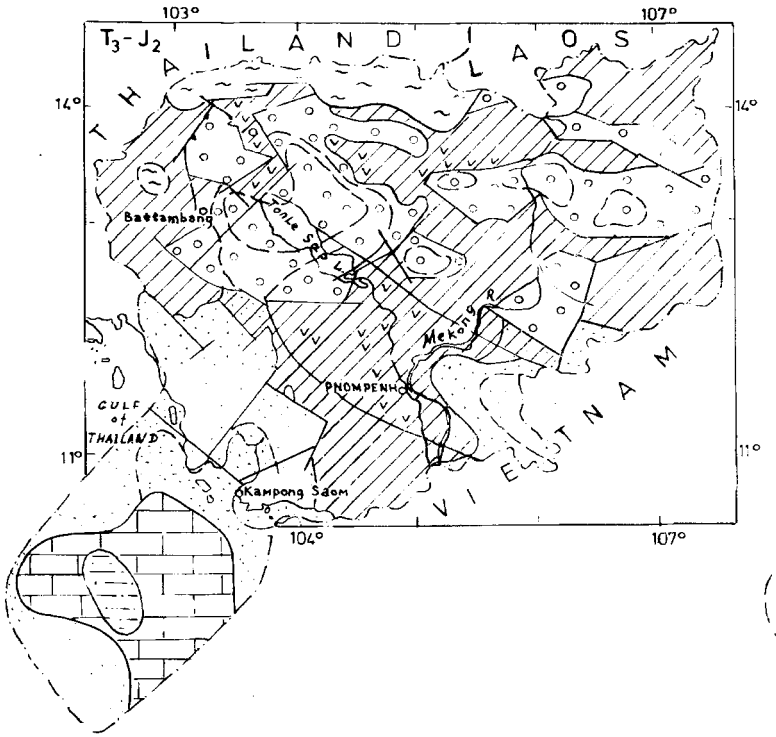


Fig. 4c. T3 – J2 structural formational complexes in Cambodia (205-160 MM yrs ago).

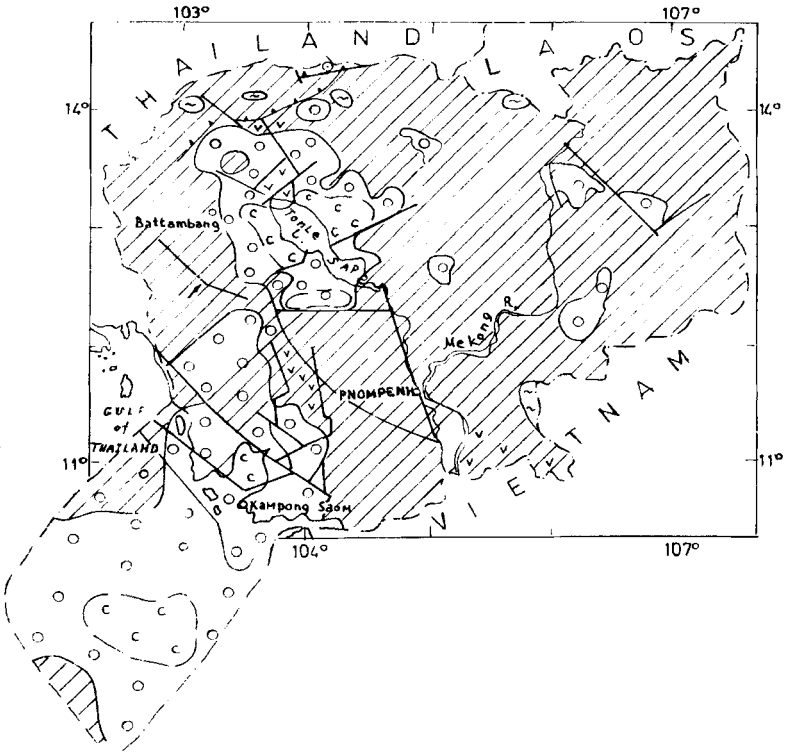


Fig. 4d. J3 – K1 structural-formational complexes in Cambodia (160-70 MM yrs ago).

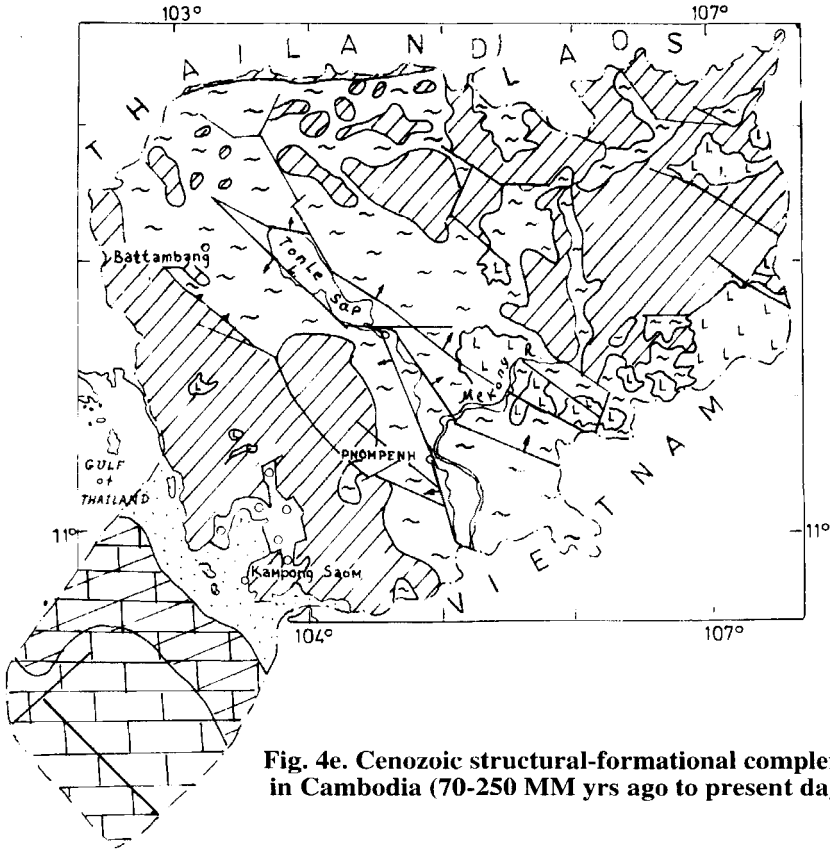


Fig. 4e. Cenozoic structural-formational complexes in Cambodia (70-250 MM yrs ago to present day).

namely: the Khorat Syneclise in the north; the intermontane Tonle Sap and Preah Troughs in the central region; and, to the south, the intermontane Kampong Saom Trough, which opens into the Gulf of Thailand, where it is overlain by the Cenozoic Siam Trough.

The territory of Cambodia is strongly faulted, and some of these faults are deep-seated. For example, the Central Cambodian Fault Zone, which runs SE across Tonle Sap Lake, forms part of the Mae Ping suture zone, which extends from Thailand through South Vietnam to the South China Sea.

POTENTIAL OIL- AND GAS-BEARING BASINS

Based on structural and tectonic features, a number of potential oil- and gas-bearing basins have been identified: these are the Siam, Khorat, Tonle Sap, Preah, Chung, and Svayrieng Basins (Fig. 6). They are discussed in turn:

Siam Basin

This basin covers more than 80,000 sq. km, of which around 60,000 sq. km are located on the Cambodian shelf of the Gulf of Thailand. In terms of structure, the basin is confined to the NE portion of the Cenozoic intermontane trough of the same name, and to the Mesozoic Kampong Saom Trough.

The basement is heterogeneous: on the shelf where Cenozoic rocks are developed, it is composed of highly-deformed Indosinian complexes; elsewhere, it is composed of deformed Palaeozoic complexes, which outcrop along the basin margin.

The sedimentary cover within the Kampong Saom Trough is composed of:

Upper Carboniferous — Lower Permian terrigenous-carbonate units, up to 1,000-m thick;

Upper Permian — Middle Triassic terrigenous-carbonate and volcaniclastic units, which reach 800-1,000 m in thickness and fill local rifts;

Upper Triassic — Middle Jurassic, mainly terrigenous units (up to 1,500-m thick), with widely- developed, transgressive silt and clay horizons;

and Upper Jurassic — Cretaceous, mostly continental formations, up to 1,500-m thick, that were deposited during the development of an intermontane depression.

Sediments in the Kampong Saom Trough exceed 3,000-4,000 m in thickness in the Kampong Saom, Bokor, Tateh, Kong and Russei Depressions (Figs. 3 and 5). The Mesozoic Kong and Russei Depressions are located on the offshore Shelf, and are separated by a NW-trending regional fault from the large area of Cenozoic subsidence which constitutes the Gulf of Thailand. Here, the sedimentary cover is composed of Indosinian orogenic complexes of Upper Triassic — Cretaceous ages. Thicknesses reach up to 3,000 m in the Pandiang Depression.

In the Rong Depression (the area of greatest subsidence on the Cambodian Shelf, which is bounded to the NE by the NW-trending Wai Uplift), the total thickness of Mesozoic-Cenozoic sediments exceeds 6,000 m, of which 1,000 - 2,000 m are Jurassic-Cretaceous in age.

Further west, the Rong Depression, located in the Central Graben of the Siam Trough, is bounded by the Naratchiwat Uplift, which separates it from the Pattani Graben, where the sedimentary succession is 8,000-9,000 m thick

Cenozoic sediments in the Siam Basin include basal continental units, which grade upwards into Palaeogene deltaic, paralic and transgressive marine units, and minor Oligocene-Quaternary carbonates.

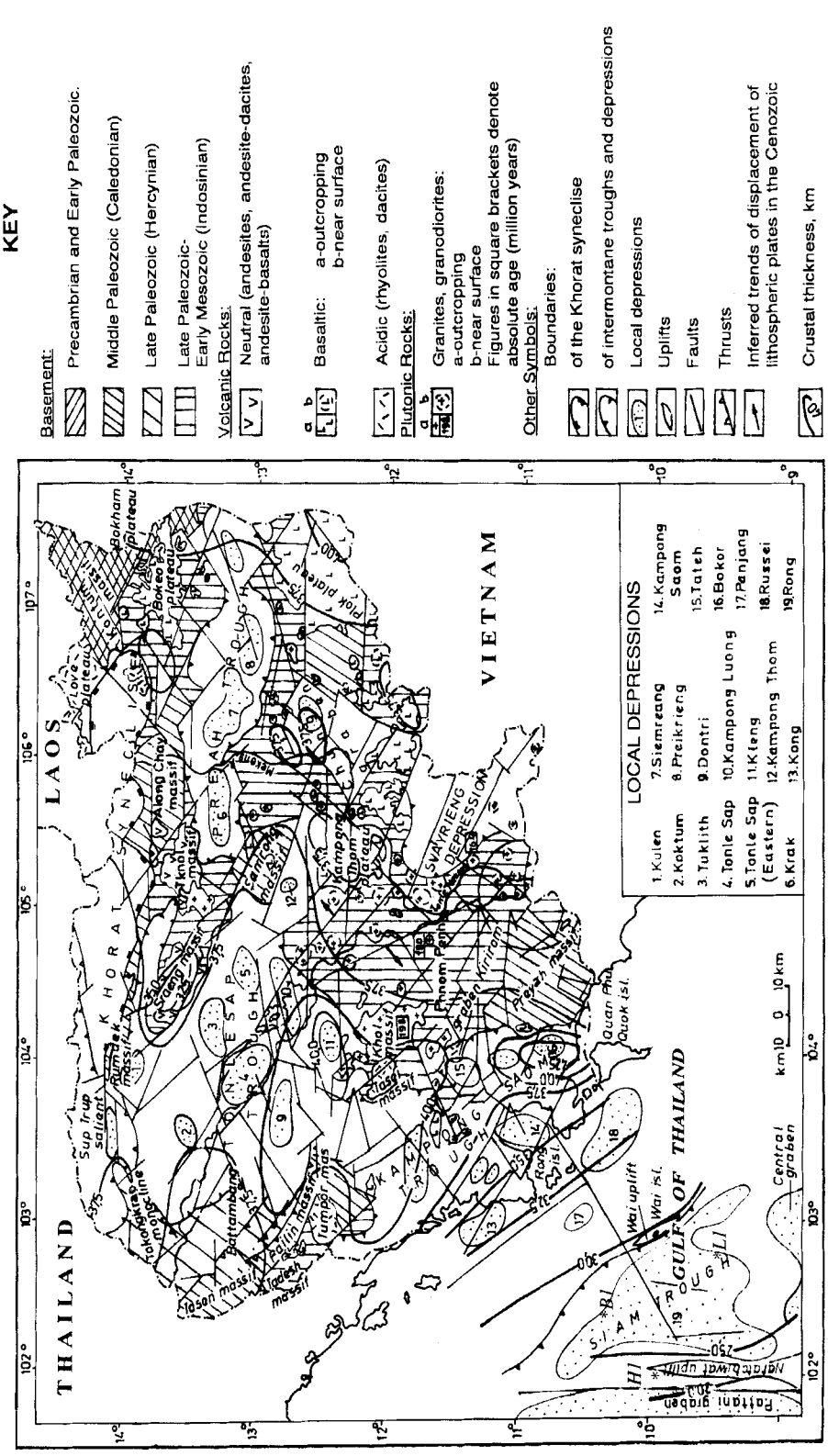
The results of oil and gas exploration in the Gulf of Thailand (offshore Thailand, Malaysia and Indonesia) indicate that about 20% of the Cenozoic succession is represented by sandy layers (up to 20-m thick). The porosity of these deltaic sandstones reaches 23%, and permeabilities 250 mD. Borehole data indicate that subsurface temperatures at depths of 2,600 m are around 126°C, reaching 176°C at 3,200 m. The geothermal gradient increases with depth from 4.8°C/1,000 m to 7.3°C/ 1,000m (below 2,600 m) (Trevenu and Clark, 1986). It is assumed that the generation of hydrocarbons in the Siam Basin is associated with the widely-developed interbedded silts and clays.

Paleogene — Middle Miocene rocks in the Siam Basin have entered the main stage of oil and gas formation (the "oil window") (Fig. 7a). In the Kampong Saom area, source-rock potential and transformation have been analysed by *Rock-Eval* pyrolysis. Temperature data indicate that Upper Jurassic-Cretaceous units have undergone early meso-catagenesis (MC1), i.e. they have just entered the main stage of oil and gas formation. Triassic-Jurassic rocks are at the same stage.

Generation of gas has probably taken place in the upper parts of the Neogene section (on the Shelf) and in the Upper Jurassic-Cretaceous section (which correspond to the upper diagenetic - early catagenetic zones). Gas and condensates may also have been generated in Mesozoic units which underlie the Cenozoic section on the Shelf, and also in Upper Carboniferous — Triassic units onshore.

In the Thai portions of the Siam Basin, drilling operations have been carried out since 1971, and the first oilfield — *Erawan* — was discovered in 1973. At the beginning of 1993, one oilfield, three oil- and gasfields and 17 gas and condensate fields had been discovered, with total initial gas reserves estimated at about 0.7 TCM (trillion cubic metres). The producing formations are Oligocene-Neogene in age, and occur at depths of 1,000-3,000 m.

During 1970-1974, a number of US and Canadian companies, together with the French Elf-Erap Co., shot 7,000 km of seismic surveys on the Cambodian Shelf, and three non-


KEY
Basement:
Precambrian and Early Paleozoic.
Middle Paleozoic (Caledonian)
Late Paleozoic (Hercynian)
Late Paleozoic-Early Mesozoic (Indosinian)
Volcanic Rocks:
Neutral (andesites, andesite-dacites, andesite-basalts)
Basaltic: a-outcropping b-near surface
Acidic (rhyolites, dacites)
Plutonic Rocks:
Granites, granodiorites: a-outcropping b-near surface
Figures in square brackets denote absolute age (million years)
Other Symbols:
Boundaries:
of the Khorat syneclise
of intermontane troughs and depressions
Local depressions
Uplifts
Faults
Thrusts
Inferred trends of displacement of lithospheric plates in the Cenozoic
Crustal thickness, km
LOCAL DEPRESSIONS
1. Kulen
2. Koktum
3. Tuklith
4. Tonle Sap
5. Tonle Sap (Eastern)
6. Krak
7. Siemreang
8. Preikriteng
9. Dontri
10. Kampong Luong
11. Kleng
12. Kampong Thom
13. Kong
14. Kampong Saom
15. Tateh
16. Bokor
17. Panjang
18. Russei
19. Rong
THAILAND
LAOS
VIETNAM
GULF OF THAILAND
KHORAT SYNECLISE
TONLE SAP DEPRESSION
SUAYRIENG DEPRESSION
Sup Trup salient
Bokham Plateau
Bokeo Plateau
Along Chow massif
Phnom Dek massif
Chhlong massif
Kravanh massif
Battambang
Tatum massif
Kirirom massif
Kirirom massif
Kirirom

producing wells were drilled, totalling 6,709 m. In 1972, *Well H-1* (lat. 9°51'19"N, long. 101°57'56"E) was drilled to a depth of 2,665 m; in 1974, *Well L-1* (lat. 9°22'23"N, long. 102°35'39"E) reached 1,875 m; and *Well B-1* (lat. 10°15'55"N, long. 102°28'13"E) reached 2,169 m (locations for H-1, L-1 and B-1 are marked on Fig. 5).

Several seismic profiles on the Cambodian Shelf were completed by the Vietsovpetro joint venture (1979, 1986), and in 1991, the Government of Cambodia announced tenders for oil gas exploration. At the time of writing, it was reported that consortia including Nawa, Premier/Persol/Santos/Ampol, Nissho Iwai, Enterprise/Sep, Technitrade Int., and Marimex were conducting seismic surveys in the Siam Basin, following licensing rounds in 1991 and 1992.

Tonle Sap Basin

This intermontane basin covers an area of 23,800 sq. km. The basement is composed of deformed and metamorphosed Palaeozoic rocks, which crop-out at the basin margins. Most of the basin is covered by Quaternary sediments, which overlie mainly terrigenous Mesozoic sediments exceeding 4,000 m in thickness. Salt-bearing strata are present within the Cretaceous interval, and may constitute good cap-rocks for hydrocarbons (Reh, 1964). By analogy with the Khorat Syneclise, a complex system of grabens may be present, and may be filled with Upper Carboniferous to Lower Permian and Upper Permian to Middle Triassic terrigenous-carbonate and volcaniclastic units, with a total thickness exceeding 1,000 m.

The "oil window" occurs between 2,000 and 3,500 m (Fig. 7b). This interval includes transgressive marine strata of Late Triassic — Jurassic ages, which may possibly be a source of liquid hydrocarbons. The Tonle Sap, Tuklith and Kleng Depressions are considered to be particularly prospective areas.

Upper Carboniferous to Middle Triassic rocks in the late catagenetic zone may have generated gas, which has migrated into reservoirs, including both overlying Mesozoic formations and a carbonate unit of Late Carboniferous to Permian age.

Khorat Basin

The Cambodian portion of this basin is located on the southern flank of the Mesozoic Khorat Syneclise (Fig. 6), and covers an area of 12,400 sq. km. Within the Thai portion of the basin, three gasfields have been discovered. Permian limestones at a depth of 3,000-4,000 m are productive.

Preah Basin

This basin, which covers 11,400 sq km, is confined to an east-west trending trough (Fig. 6); up to 3,000 m of Mesozoic sediments in the Preikrieng, Siemreang and Krak depressions (see Fig. 5 for locations). The basement is composed of folded Indosinian rocks. Triassic to Middle Jurassic units have passed through the "oil window", and may well constitute source rocks for hydrocarbons.

Chung Basin

With an area of 2,600 sq km, this basin is a part of a north-south trending graben in the Indosinian foldbelt (Fig. 5), and is filled by Mesozoic and, probably, Cenozoic sediments more than 3,000-m thick.

Svayrieng Basin

This intermontane basin in the SE of Cambodia (Figs 5 and 6) is filled by around 3,000 m of Mesozoic sediments, and covers an area of 6,700 sq. km.

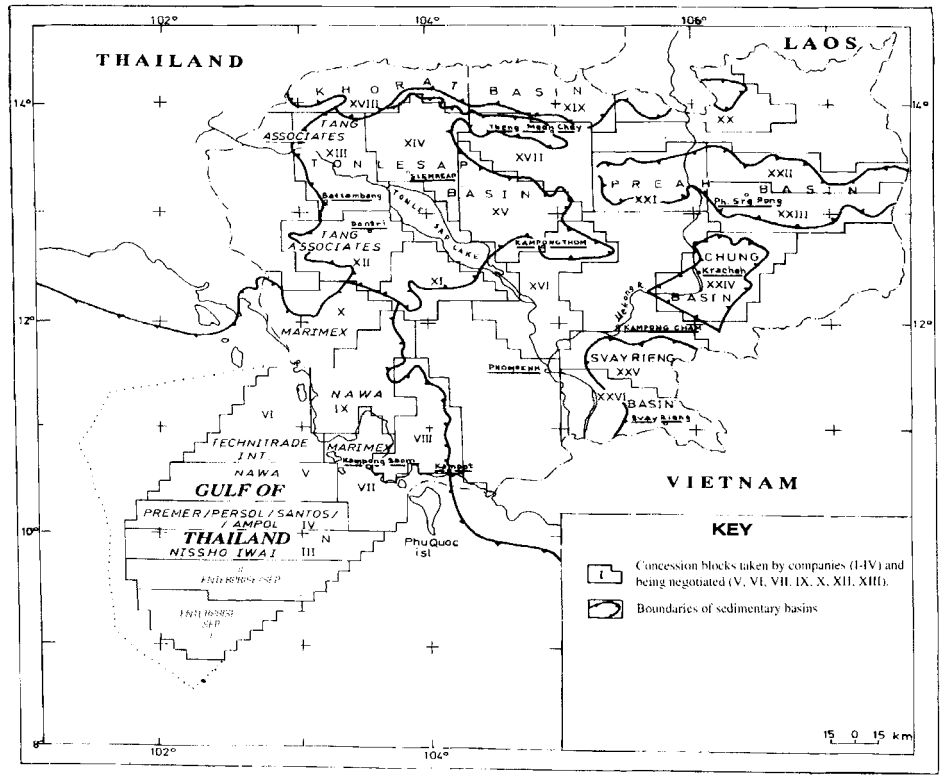


Fig. 6. Location map of the potentially-productive sedimentary basins and concession blocks on - and offshore Cambodia.

PETROLEUM POTENTIAL OF CAMBODIAN BASINS

The most favourable geological conditions for hydrocarbons accumulations occur in the Siam Basin, including the offshore Shelf and parts of the Gulf of Kampong Saom. Similar conditions probably exist in the central part of the Tonle Sap Basin.

Cenozoic, Mesozoic and Upper Palaeozoic successions are considered to be promising for hydrocarbons exploration. In most of the basins, Upper Triassic-Jurassic units have probably passed through the main stages of oil and gas formation, and are capable of generating both liquid and gaseous hydrocarbons.

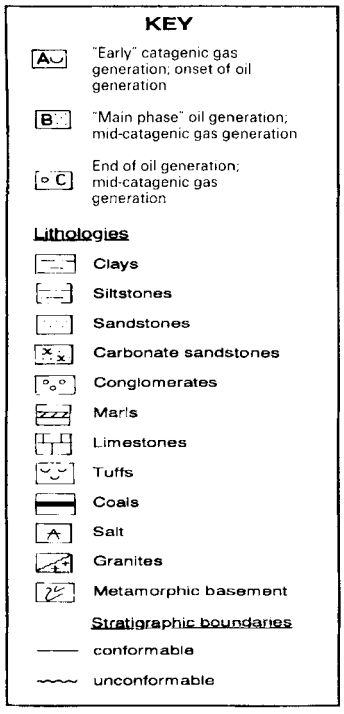
The Late Permian to Middle Triassic rift complex is probably also a source of oil and gas, but is only present in a limited area. The underlying Carboniferous — Permian units may have generated (and contain) predominantly gas and condensate accumulations. Upper Jurassic — Cretaceous units may also have generated minor volumes of gas.

On the Shelf, the main generating and reservoir formations are Oligocene-Miocene in age.

OIL AND GAS RESOURCES

Due to the limited nature of the geological and geophysical coverage, estimates of total hydrocarbons resources have been made for the entire Mesozoic and Cenozoic sedimentary cover, and total around one billion (10^9) tons of oil-equivalent.

The average volumetric density of the "in-place" hydrocarbons resources offshore amounts to 5,900 t.o.e./cu. km (tons oil-equivalent per cubic kilometre), with minimum



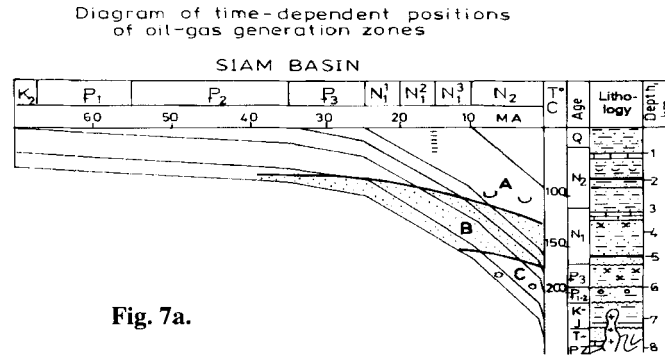


Fig. 7a.

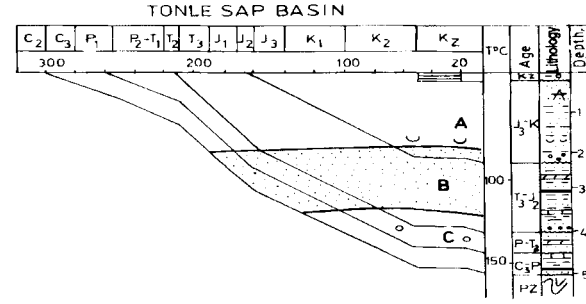


Fig. 7b.

Fig. 7. Burial history plots for (a) the Siam Basin; (b) the Tonle Sap Basin.

and maximum values of 2,100 and 8,700 t.o.e./cu. km, respectively. Onshore, the average volumetric density is 1.5 times lower than this, with a range of values from 60 to 6,400 t.o.e./cu. km. More than 30% of the initial "in-place" hydrocarbons resources are concentrated in the onshore basins of Cambodia.

SUMMARY AND CONCLUSION

Most of Cambodia's on- and offshore territory is considered to be potentially oil- and gas-bearing. The Tonle Sap, Preah and Chung Basins, together with the northern flank of the Siam Basin, and the southern flank of the Khorat Basin, are considered to be important new exploration targets in SE Asia.

The Upper Palaeozoic — Cenozoic sedimentary succession includes source, reservoir and cap rocks. Carboniferous and Middle Triassic strata are predominantly gas-generating. The Upper Triassic — Middle Jurassic sediments that have passed through the "main phase" of oil and gas generation in the Siam (onshore), Tonle Sap and parts of the Preah and Chung Basins are capable of generating both liquid and gaseous hydrocarbons. In the Siam Basin, the "oil window" encompasses Oligocene-Miocene strata, which account for the basin's potential.

Reservoir horizons are dispersed through almost the entire sedimentary column. Regional, but not always effective and persistent cap-rocks may be formed by Mesozoic and Cenozoic clay horizons and, locally, by salt horizons (e.g. in the Tonle Sap Basin).

Numerous uplifts have been identified by structural analyses and remotely-sensed imagery, and these may serve as traps for hydrocarbons. Most of the uplifts located onshore are associated with Indosinian movements. On the Shelf of the Gulf of Thailand, the main stage of structural deformation occurred at the Upper Miocene-Pliocene boundary. Some of the structures are formed on basement uplifts. Folding of marginal parts of the troughs may have been due to lateral displacement of Paleozoic blocks over the thrust system. Individual domes are associated with salt tectonics.

The total volume of combined hydrocarbon resources of Cambodia is estimated to be one billion t.o.e., and on the basis of World average recovery factors (31.5% for oil and dissolved gas, 65% for condensates, and 80% for free gas), recoverable hydrocarbons are estimated to total 0.6 — 0.8 B t.o.e.

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